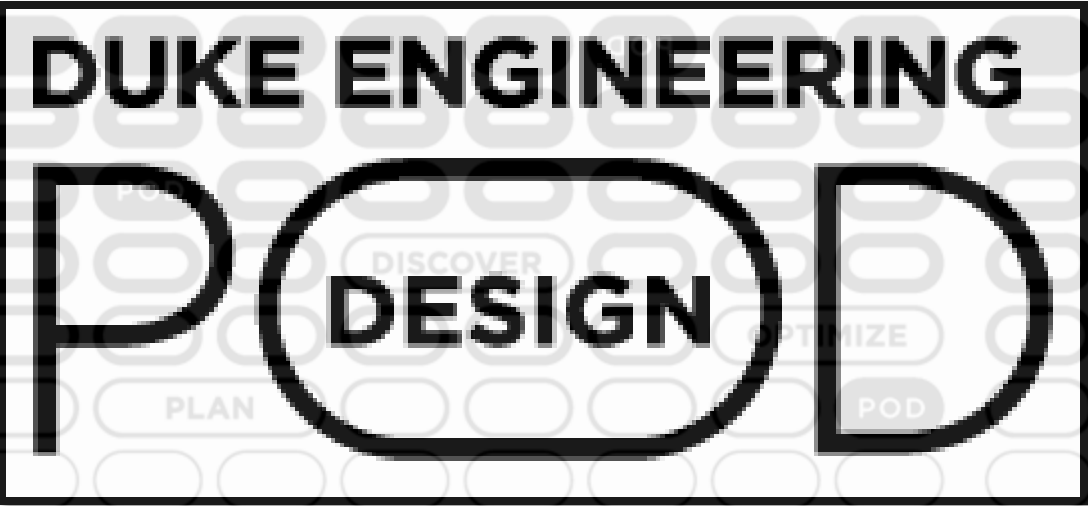


Adaptive Motion Stabilization Device for Spasticity

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Design Problem & Motivation

Who:

Student “A” is a 7-year-old with dystonia (a condition that causes involuntary muscle contractions)

What:

Create an assistive arm device that:

1. Supports Student ”A”’s left arm during the school day
2. Does not restrict his range of motion
3. Reduces fatigue

Why:

Increase Student ”A”’s classroom participation, confidence, and independence

Condition Background

Cause of Dystonia: Abnormal functioning of brain regions responsible for movement control. ¹

Symptoms: Uncontrolled muscle contractions that intensify with increase stress/fatigue/physical activity—results in less stable movement and requires constant supervision and support. ²

Design Criteria and Testing

Design Objective	Target Values	Test Description	Results
Safety	1. 100% Latex- free, lead-free paints, and phthalate-free plastics 2. All edge radius > .17 in 3. Insertion depth < 2.25x hole diameter 4. Wires are fully enclosed	Test material + structural safety, and electrical safety (pass/fail). Safety is achieved if 80% of the tests pass.	1-3: <i>PASS</i> 4: TBD
Cost	Total cost ≤ \$100	Sum material cost → pass/fail	<i>FAIL</i> × <i>Total cost: \$128.8</i>
Range of Motion	Left arm range of motion (ROM): Elbow: flexion 0–150° Shoulder: flexion 0–180° Rotation 0–90° Device must enable 80% of desired ROM	Measure elbow and shoulder range of motion on 10 participants	TBD
Comfort	Average rating ≥ 2.50/3.00 on UDS survey	Surveyed 20 participants who simulated user experience in the device	<i>PASS</i> ✓ <i>Average rating: 2.85/3.00</i>
Adaptability	Range of forearm + elbow adjustment must be > 1.25cm (Fig. 3).	Use vernier calipers to forearm + elbow extension, must be > ±1.25cm.	<i>PASS</i> ✓ <i>Can be adjusted + 6.35cm</i>
Surface Independence	1. Typing speed of 1/3 of user's baseline speed or more 2. 80% accuracy	20 users will attempt to achieve target value at 5 different heighted desks	TBD

Final Design Overview

This device supports Student "A" through elbow flexion, shoulder internal/external rotation, and shoulder flexion. It uses metal telescoping poles for lateral movement (Fig. 3), 3D-printed plastic joints (Fig. 4, Fig. 6), as well as a bearing for yaw movement (Fig. 5). For added comfort, the device uses foam padding and leverages an IMU-Servo combination for stabilization (Fig. 7). The device is adjustable in size to fit as the client grows (Fig. 2).

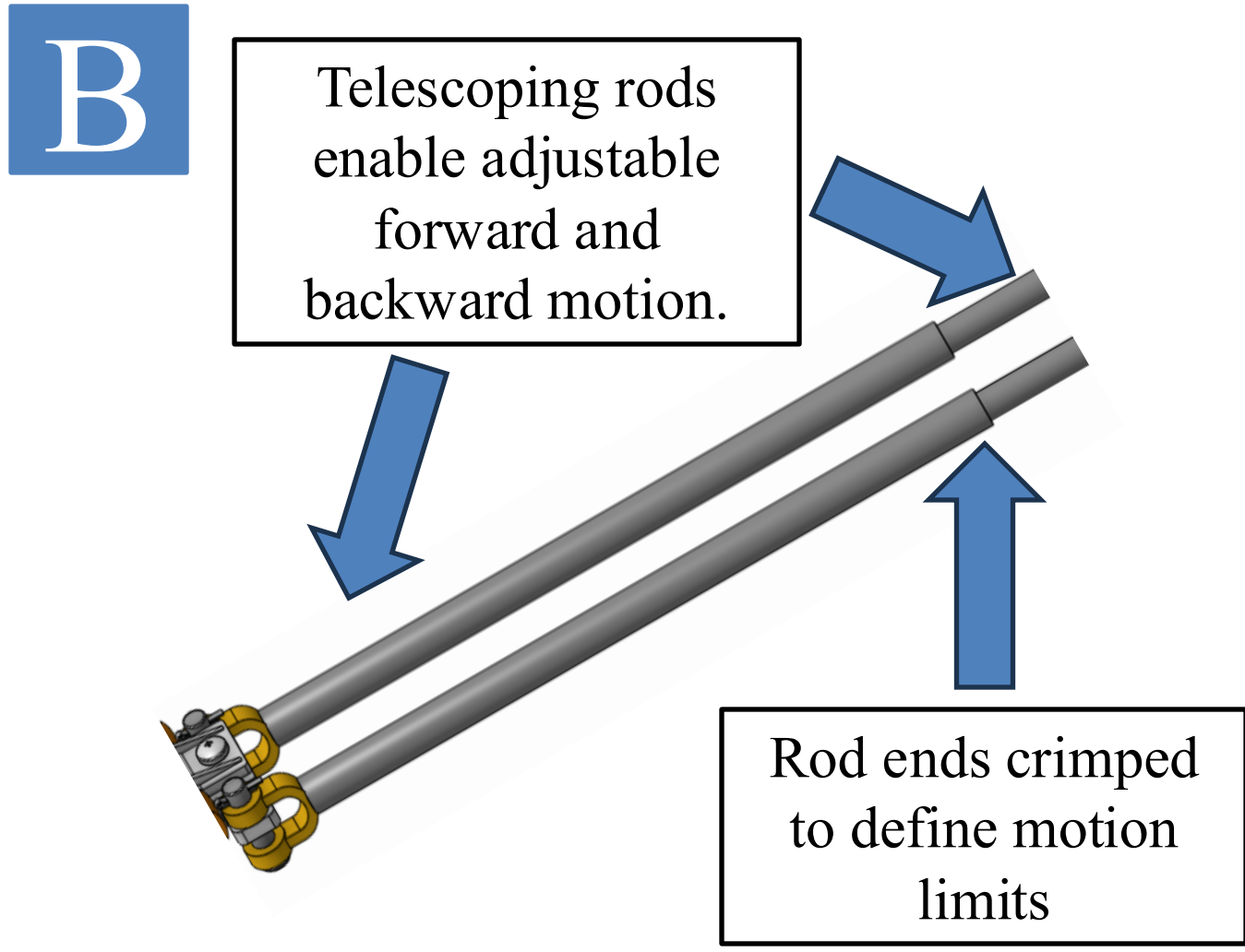


Fig. 3: Telescoping Poles for Forward and Backwards Motion

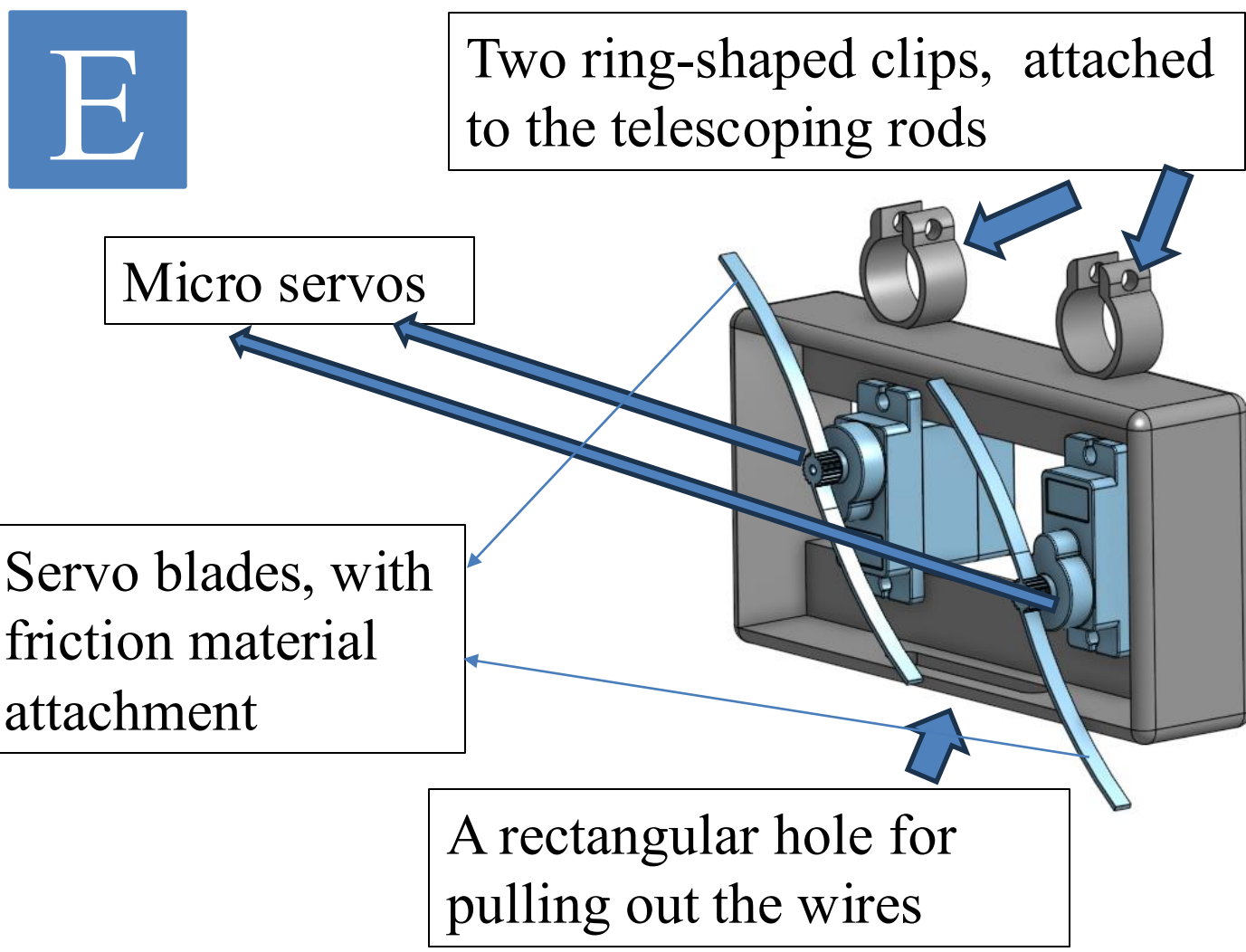


Fig. 6: 3D Printed Servo Housing

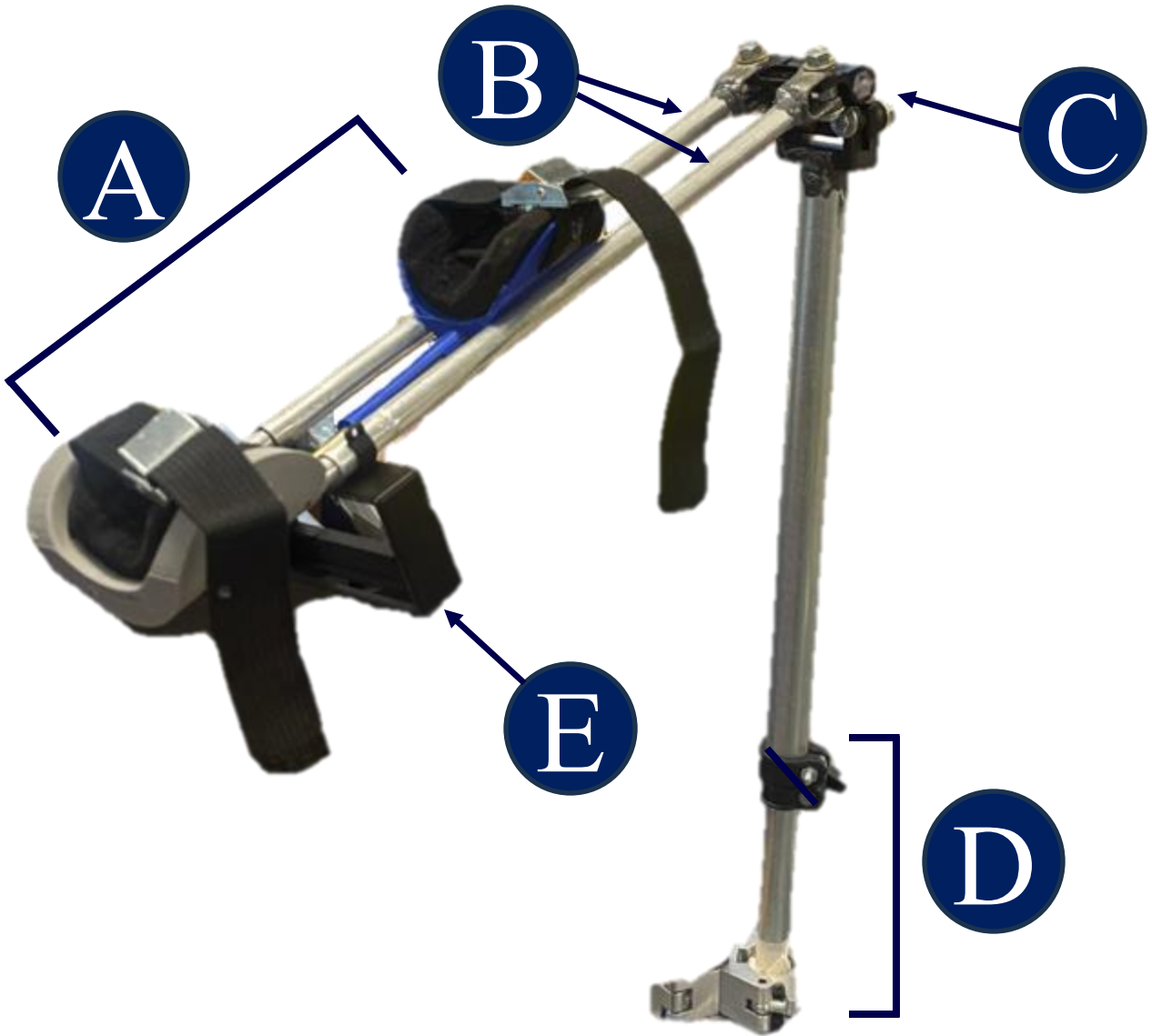


Fig. 1: Full Assistive Arm Device; labels corresponding to Figs. 2-7

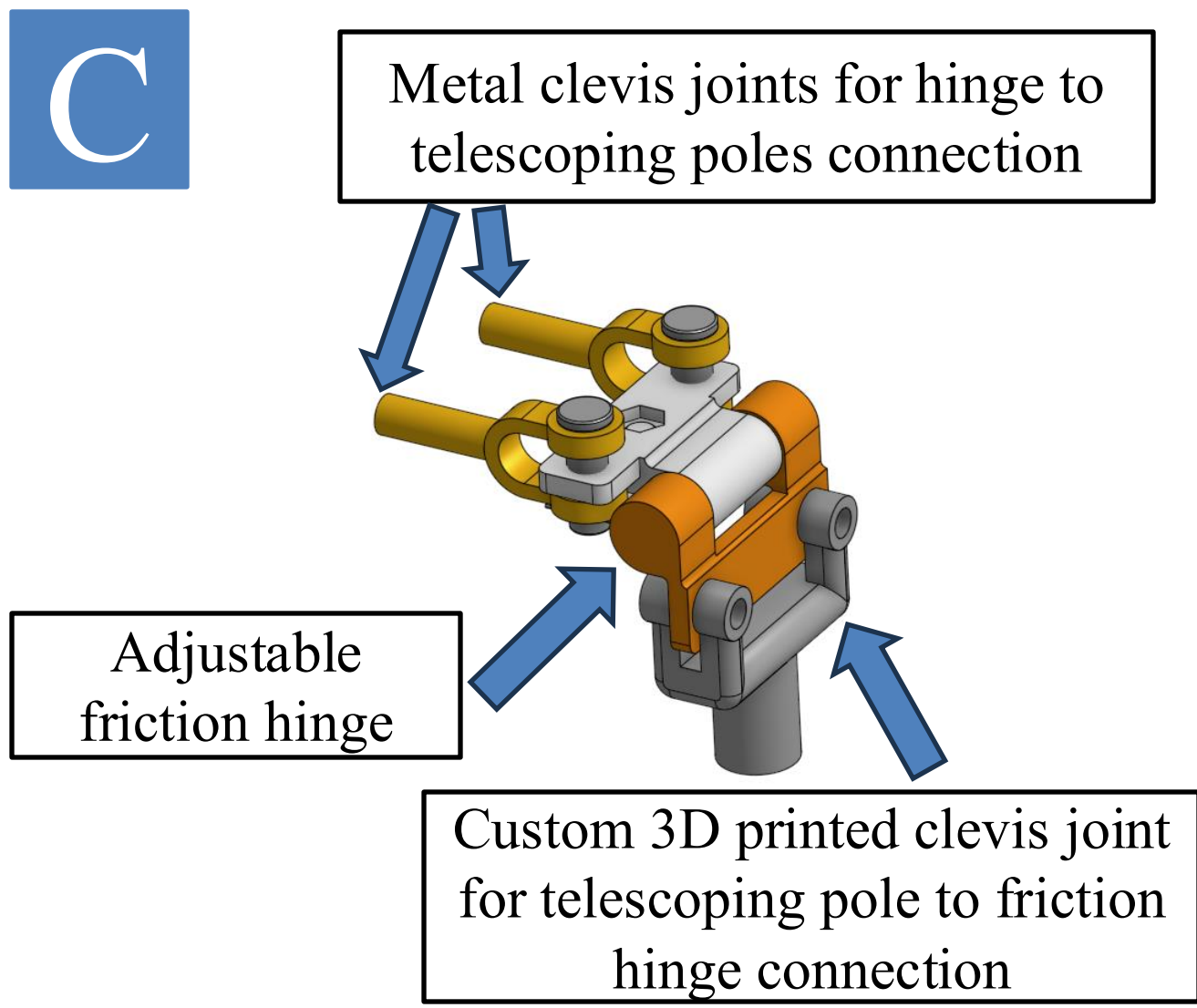


Fig. 4: Friction Hinge with Clevis Joints for Telescoping Pole Attachment

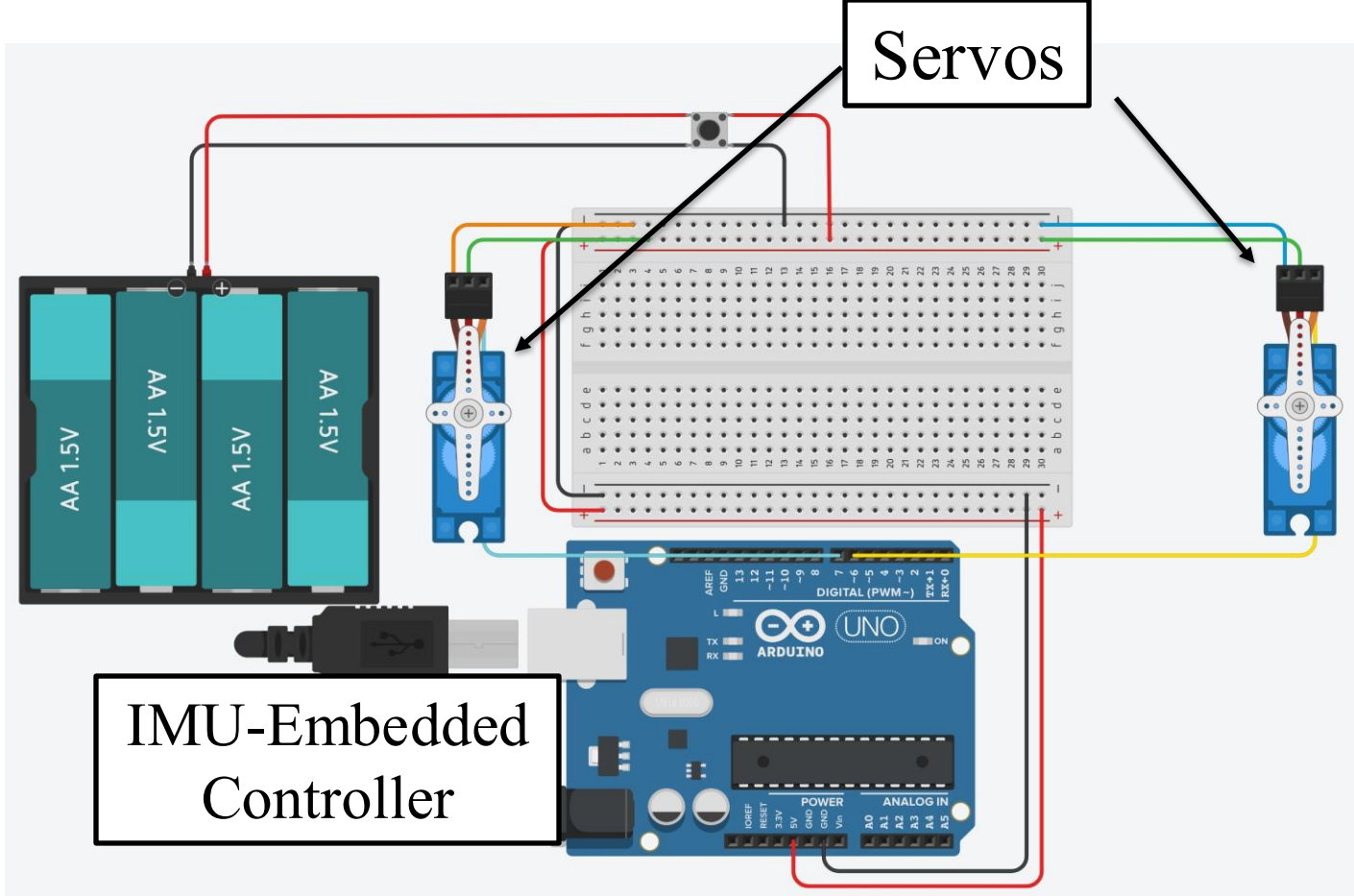


Fig. 7: Circuit Diagram with IMU-Integrated Microcontroller Powering 2 Servos

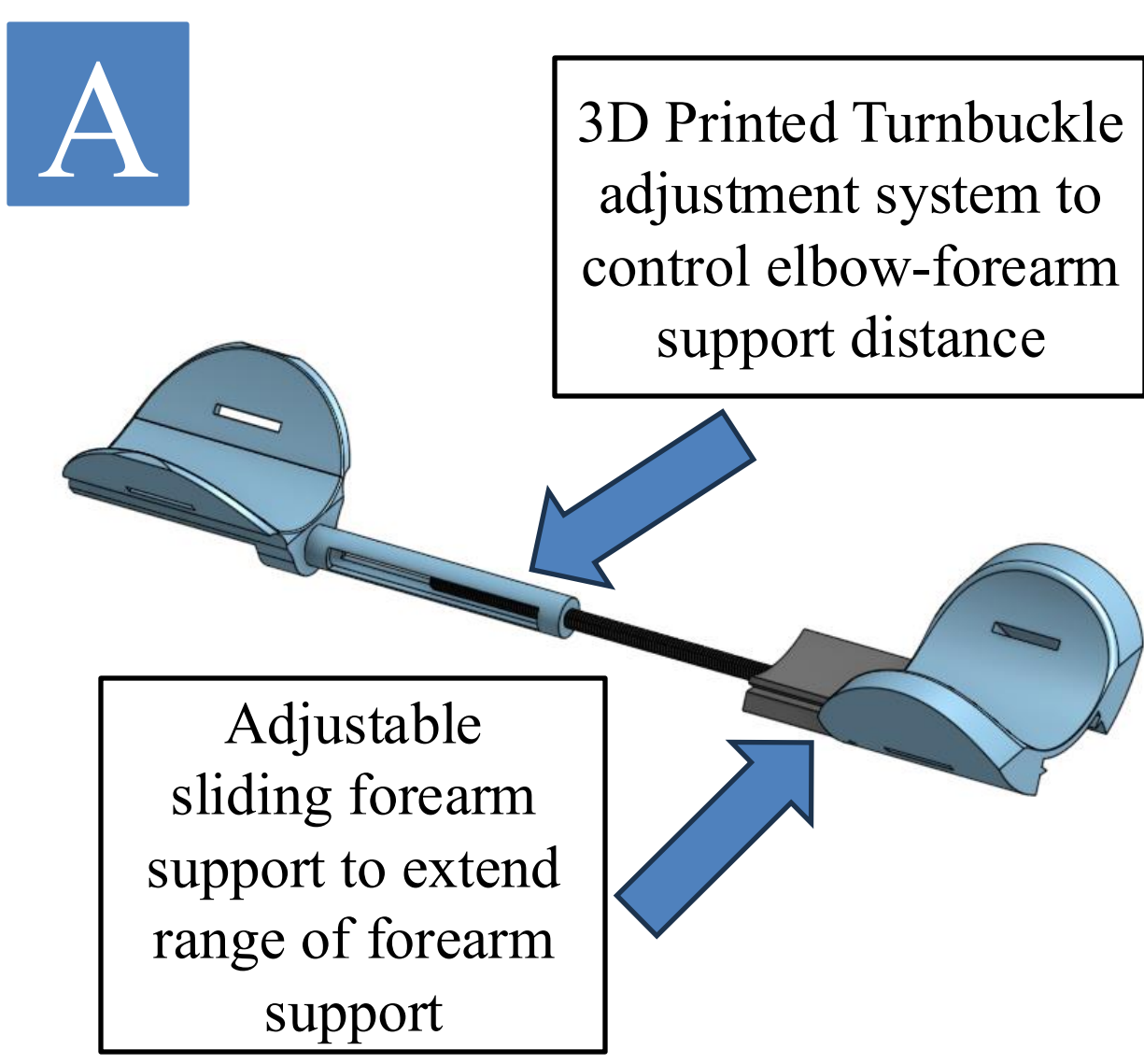


Fig. 2: Forearm and Elbow Supports with Extending Parts and Turnbuckle System

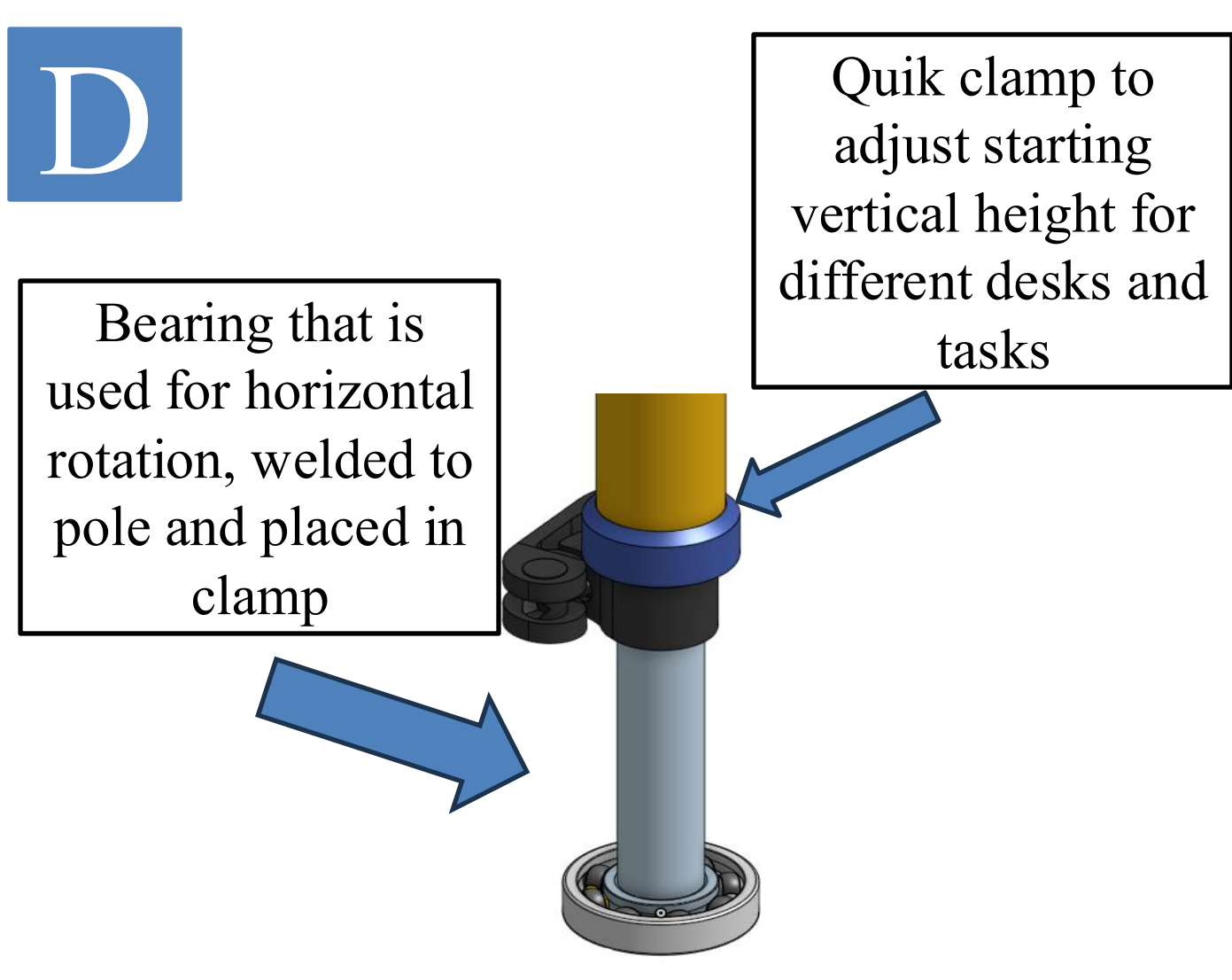


Fig. 5: Bearing and Vertical Pole Attachment for Yaw Movement

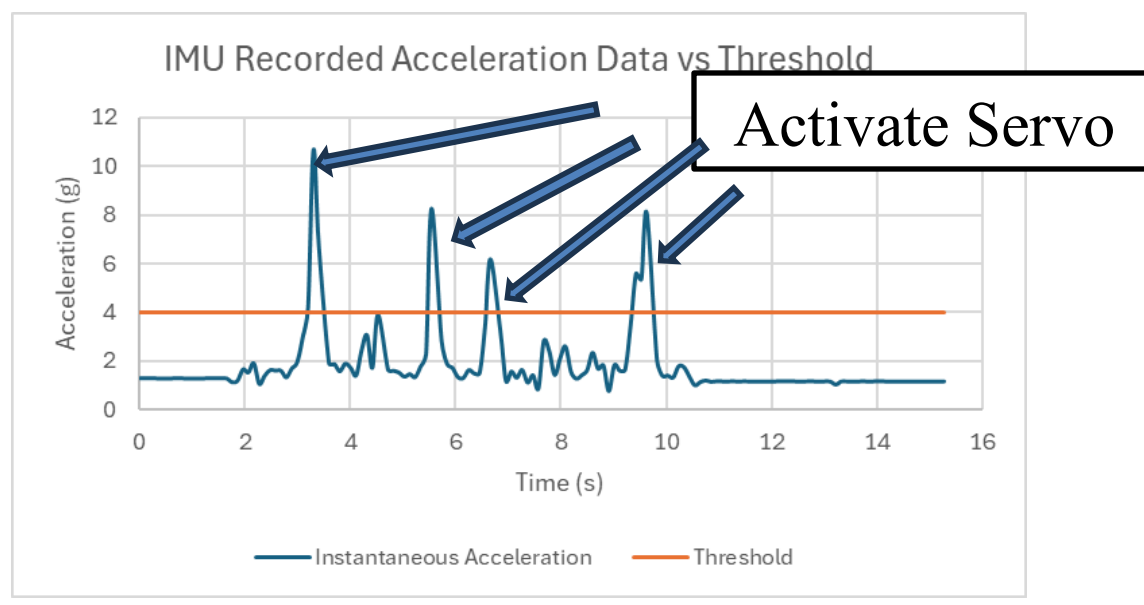


Fig. 8: Graph Showing Servo Activation During Acceleration Spikes

Conclusion & Future Work

Our team successfully designed and assembled an adjustable, comfortable arm support device featuring a Servo–IMU stabilization system, promising potential for improving user independence in classroom activities.

To further develop and improve this solution, we could:

- Create new attachments for increased surface-independence and use on various surfaces
- Optimize servo control and IMU filtering for smoother active stabilization, leveraging Kalman Filters
- Explore lighter materials, instead of heavy metal tubes, to reduce total device weight
- Explore alternative designs that could be used for a broader audience instead of just for Student "A"

Acknowledgements

Our team would like to thank the following for making this project possible:

Dr. Sonia Bansal, Professor
Dr. Roarke Horstmeyer, Professor
Dagan Trnka, POD Manager
Nicole Ratterman, Teaching Assistant
Ginny Poarch, Client
Maenette Wahrenburg, Client Assistant

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